Vibhu Gautam

Generative AI, Agentic AI & Data Delivery Leader

Paonta Sahib, India | +91-9317860077 | vibhu.gautam7@gmail.com | linkedin.com/in/vibhugautam7

EXECUTIVE PROFILE

Generative AI, agentic AI, and data delivery leader with 12+ years of global experience across **Nestlé, McKinsey, BCG, Micron, University of York**, and higher education. Translates complex business problems into scalable AI-enabled products, automation, data platforms, and decision systems. Combines Python engineering, machine learning, MLOps, cloud deployment, retrieval-augmented generation, and agentic workflows with data governance, executive stakeholder alignment, and practical technical enablement.

CORE COMPETENCIES

Generative & Agentic AI:	LLM application design, prompt engineering, retrieval-augmented generation, embeddings, vector databases, tool-using agents, evaluation, and human-in-the-loop workflows.
Applied AI & ML:	Predictive modeling, causal ML, Bayesian statistics, hypothesis testing, uncertainty quantification, model assurance, and responsible AI practices.
AI Systems & MLOps:	Docker, Kubernetes, FastAPI, CI/CD, API integration, model deployment, evaluation workflows, monitoring, and reproducible delivery.
Data Platforms & Cloud:	Snowflake, Azure (ADF, Synapse, ADLS, APIM), AWS, Databricks, PySpark, Spark, REST APIs, and data quality controls.
Programming & Product Build:	Python (Pandas, NumPy, Scikit-learn, TensorFlow), SQL, Streamlit, Power BI, Tableau, Excel/VBA, and project-based prototyping.
Delivery & Enablement:	AI use-case discovery, technical training, stakeholder alignment, agile product lifecycle, curriculum design, and executive communication.

ACADEMIC LEADERSHIP & INDUSTRY TEACHING

Visiting Faculty – Data Science, Python & Applied AI Jul 2026 – Present

Indian Institute of Management Sirmaur (IIM Sirmaur) *Sirmaur, India*

- Teach postgraduate management cohorts in Data Science, Business Analytics, and programming fundamentals using Python.
- Deliver applied modules in Generative AI and agentic AI, connecting prompting, knowledge retrieval, tool-using workflows, and responsible adoption to managerial decision-making.
- Bridge academic frameworks with contemporary industry practice in data architecture, AI-enabled decision systems, and executive communication.

Professor of Practice – Computer Science Aug 2024 – Present

UPES, School of Computer Science *Dehradun, India*

- Lead curriculum design and delivery for applied courses in Data Architecture, Snowflake, Big Data, MLOps, and Applied Analytics.
- Built and deployed the public **People Analytics Lab** platform, connecting workforce simulation and diagnostics to live business cases.
- Mentor students on production analytics architectures, end-to-end cloud pipelines, automated CI/CD, and AI-driven applications.

GLOBAL INDUSTRY EXPERIENCE

Innovation Lead – Data & AI Mar 2023 – Aug 2024

Nestlé Global Headquarters *Gurugram, India*

- Spearheaded global data and AI products for complex enterprise decision-making, including workforce intelligence and operational insight use cases.
- Architected scalable Azure-based analytics platform (ADF, Synapse, ADLS), establishing standardized global HR data models and REST APIs.
- Implemented automated data governance, validation checks, and model assurance practices, improving delivery reliability across markets.
- Translated ML model outputs into actionable executive narratives, enabling proactive mitigation of organizational flight risk.

Data Scientist – Full Stack Data Products Aug 2021 – Apr 2023

Nestlé UK *York, United Kingdom*

- Developed end-to-end analytics products spanning ingestion, feature stores, predictive modeling, APIs, and Power BI dashboards.
- Built predictive machine learning models for employee attrition risk, workload pressure, and retention drivers for proactive HR intervention.
- Delivered interactive self-service dashboards and analytical workflows supporting enterprise workforce planning and employee engagement.
- Coordinated cross-functional teams across the UK and India, standardizing HR metrics and reporting into decision-support products.

GLOBAL INDUSTRY EXPERIENCE (CONTINUED)

Marie-Curie Early Stage Researcher – Data Statistics & Decision Systems May 2019 – Aug 2021
University of York *York, United Kingdom*

- Developed Bayesian statistical models and uncertainty-aware methods for model assurance, reliability analysis, and responsible AI.
- Researched predictive confidence and safety-critical decision systems within an EU Horizon 2020 international research consortium.
- Published peer-reviewed research papers in international venues including **SafeAI 2021** (AAAI Spring Symposium) and **EAI 2021**.

Data Scientist – Predictive Analytics & Data Platforms Aug 2017 – Nov 2018
Micron Technology *Hiroshima, Japan*

- Developed distributed Spark/PySpark data pipelines and classification models for real-time equipment fault detection and yield optimization.
- Integrated semiconductor fabrication telemetry with enterprise manufacturing datasets to identify throughput patterns and process bottlenecks.
- Conducted model validation, feature importance analysis, and distribution checks to ensure production reliability in automated manufacturing.

Analyst – Practice Analytics & Benchmarking Apr 2015 – Aug 2016
Boston Consulting Group (BCG) *Gurugram, India*

- Maintained proprietary benchmarking databases, financial formulas, and market estimation models supporting global consulting casework.
- Automated recurring client-ready analytical outputs using reusable calculation logic, improving delivery turnaround and scalability.
- Collaborated with consulting teams across industries to analyze workforce productivity, cost baselines, and operating models.

Analyst – Corporate & Business Function Benchmarking Nov 2012 – Mar 2015
McKinsey & Company *Gurugram, India*

- Conducted multi-function benchmarking and organizational diagnostics across HR, Finance, IT, Legal, Procurement, and Marketing.
- Developed FTE allocation models, spans-and-layers diagnostics, and shared-services assessments for global restructuring and M&A synergy.
- Evaluated location-cost dynamics, functional productivity, and operating-model options to identify cost optimization and centralization levers.

VBA / Analytics Automation Developer Jul 2011 – Jul 2012
Brochet *Oxfordshire, United Kingdom*

- Engineered Excel/VBA, VSTO, XML, and SQL Server reporting automation, eliminating manual workflows and standardizing client delivery.

SELECTED GENAI AND DATA PROJECTS

Jihokai Learning Platform: Role-aware learning-platform architecture with Firebase authentication and Google Apps Script / Sheets workflows for courses, quizzes, and progress.

Agentic Workflow Prototypes: Tool-using workflow patterns combining structured prompts, retrieval, context management, human review, and operational hand-off.

Applied Decision Intelligence: Causal ML and uncertainty-aware modelling for decision support, scenario analysis, validation, and responsible use.

Enterprise GenAI Enablement: Applied curriculum covering prompting, RAG concepts, knowledge grounding, agentic workflows, evaluation, security, and use-case development.

AI Deployment Toolkit: Practical patterns for Python, FastAPI, Streamlit, Docker, Kubernetes, cloud deployment, CI/CD, and scalable data integration.

People Analytics Lab: Domain application of analytics and simulation for workforce diagnostics, skills intelligence, and organisational decision support.

EDUCATION & PROFESSIONAL CREDENTIALS

Professional Certificate in Data Science – University of Oxford, UK 2017
M.Sc. in Computer Science – University of Southampton, UK 2009 – 2010
B.Tech. in Electronics & Communication Engineering – Chitkara University, India 2005 – 2009

RESEARCH INTERESTS, PUBLICATIONS & CERTIFICATIONS

Research Focus: Generative AI in Managerial Decision-Making; Human–AI Collaboration; Decision Intelligence under Uncertainty; Responsible AI & Governance; Organizational Behavior in Automated Workplaces.

Selected Publications: Peer-reviewed papers on uncertainty-aware decision systems and conversational AI published in **SafeAI 2021** (AAAI Spring Symposium) and **EAI 2021**.

Certifications: Machine Learning & AI (upGrad, 2019) | Data Science for IoT (University of Oxford, 2017).